

CHAPTER

49



Sexual Dysfunction and Infertility

Learning Objectives

- ⇒ The description of the terms sexual dysfunction and infertility
- ⇒ To order the interpret relevant investigations of sexual dysfunction and infertility
- ⇒ To know the description of differential diagnosis of sexual dysfunction and infertility

Introduction

Sexual dysfunction and infertility are two different symptoms that can disturb the emotional or mental peace of the couple; each of which are having distinctive implications. Sexual dysfunction may disturb the couple only; whereas presenting complaint of infertility may disturb the couple in association with people who are associated with these couple and thus leading to the social disturbances too. Earlier, discussion in related to sexual dysfunction and infertility were considered to be a taboo, but now because of excess information available in the social media, internet and peer influences.



Figure 217: Sexual dysfunction

There are two terms used in the field of infertility, infertility and sterility. Both literally mean inability to conceive after one year of sexual life without contraception when the couple are anxious to have progeny. Some distinguish the two terms with slight difference. Sterility refers to the absolute state of inability to conceive while infertility refers to the relative state of failure to conceive. Infertility can be either primary, where the conception has never occurred or secondary, where conception has failed to occur after a period of fertility.

Sexual Dysfunction

Male sexual function involves

1. Intact libido (sexual desire)
2. Ability to achieve and maintain erection
3. Ejaculation and orgasm
4. Detumescence

Male sexual dysfunction (libido, erection, ejaculation, orgasm and detumescence) interferes in or ability to engage in sexual intercourse. We are going to analyse each of these parameters.

- ⊗ **Libido:** This is the conscious component of sexual function. Decreased libido manifests as a lack of sexual interest or a decrease in frequency and intensity of sexual thoughts, either spontaneous or in response to erotic stimuli. Libido is sensitive to testosterone levels as well as to general nutrition, health and drugs. Conditions particularly likely to decrease libido include hypogonadism, uremia and depression. Antidepressants and antipsychotics are likely to produce reduced libido.

- ⊗ **Erection:** Erection occurs as the result of a complex neuropsychological process. Higher cortical input and a sacral mediated parasympathetic reflex are combine to stimulate erection. Nerve output signals through the pudendal nerves, which traverse the posterolateral aspect of the prostate. Terminating in penis, these nonadrenergic, noncholinergic nerves activate nitric oxide synthase, producing nitric oxide, which relaxes smooth muscle lining the sinusoidal spaces that connect the arterioles and venules within the corpora cavernosa. The blood flow within the sinusoids increases markedly, distending them and compressing the venules, causing veno occlusion. The increase inflow and veno-occlusion together produces penile rigidity.
- ⊗ **Ejaculation and orgasm:** Ejaculation is controlled by the sympathetic nervous system, leading to contraction of epididymis, vas deferens, prostate, muscles of pelvic floor. In addition, the neck of the bladder closes, preventing retrograde ejaculation of semen into the bladder. Orgasm is the highly pleasurable sensation that occurs in the brain generally simultaneously with ejaculation.
- ⊗ **Detumescence:** Restoration of size of the penis to the normal size after ejaculation and orgasm.

Erectile Dysfunction

Erectile dysfunction is the inability to attain or sustain an erection satisfactory for sexual intercourse. Most erectile dysfunction is related to vascular, neurological, psychological and hormonal disorders. Drug use can also be a cause for ED. Evaluation typically includes screening for underlying disorders and measuring testosterone levels. Earlier, the term impotence was used, which was replaced by erectile dysfunction.

- ⊗ Primary ED (i.e. man has never been able to attain or sustain erections) is rare and is almost always due to psychological causes (guilt, fear of intimacy, depression or severe anxiety).
- ⊗ Most often, ED is secondary, (i.e. a man who previously could attain and sustain erections, but no longer can). Over 80% of secondary ED cases have organic pathology. Psychological factors must be considered in every case.
- ⊗ Major organic causes of secondary ED are vascular (penile atherosclerosis) and neurological (stroke, partial complex seizures, multiple sclerosis, peripheral neuropathy, autonomic nervous dysfunction) disorders, often stemming from atherosclerosis and diabetes mellitus; complications of surgery, prostate surgery in particular; hormonal disorders (hypogonadism); drugs (excess alcohol consumption) and structural diseases of penis (priapism).
- ⊗ Psychological causes of ED include performance

anxiety, stress or mood disorder (depression in particular).

- ⊗ Diagnosis is done by clinical evaluation, screening for depression and assessing the level of testosterone in the patient

Female Sexual Dysfunction

Female sexual function involves

1. Intact libido (sexual desire)
2. Sexual arousal due to environment and partner
3. Vaso congestion and lubrication of vagina and external genitalia
4. Orgasm

Female sexual dysfunction involves disturbances in one or more of above said parameters. We will look into each of the parameters.

- ⊗ **Intact libido:** Brain areas involved in cognition, emotion, motivation and organization of genital congestion are all activated. Neurotransmitters acting on specific receptors are involved. Based on known actions of drugs and on animal studies, some neurotransmitters appear to be prosexual, they include dopamine, norepinephrine, etc.
- ⊗ **Sexual arousal due to environment and partner:** Privacy for the couple, surroundings of the environment where the couple reside and foreplays done by sexual partner makes the female to get sexually aroused.
- ⊗ **Genital congestion:** This reflexive, autonomic response occurs within seconds of an erotic stimulus and causes genital engorgement and lubrication. Smooth muscle cells around blood spaces in the vulva, clitoris, and vaginal arterioles dilate, increasing blood flow (engorgement) and transudation of interstitial fluid across vaginal epithelium. As women age, basal genital blood flow decreases, but genital congestion in response to erotic stimuli may not.
- ⊗ **Orgasm:** Peak excitement occurs, characterized by contractions of pelvic muscles every 0.8 second, and is followed by slow release of genital congestion.
- ⊗ Female sexual dysfunction may involve decreased or increased sexual responsiveness. Classification is determined by symptoms. There are 5 major categories of decreased sexual responsiveness and one of increased sexual responsiveness (persistent genital arousal disorder).
- ⊗ **Sexual desire/ interest disorder:** is absence of or decrease in sexual interest, desire, sexual thoughts and fantasies and an absence of responsive desire.
- ⊗ **Sexual arousal disorder:** is lack of subjective genital arousal or both.
- ⊗ **Orgasmic disorder:** involves orgasm that is absent, markedly diminished in intensity, or markedly

delayed in response to stimulation despite high levels of subjective arousal.

- ⊗ **Vaginismus:** reflexive tightening around vagina when vaginal entry is attempted or completed despite woman's expressed desire for penetration and when no structural or other physical abnormalities are present.
- ⊗ **Dyspareunia:** This is pain during attempted or completed vaginal penetration or intercourse. Provoked vestibulodynia (formerly called as vulvar vestibulitis), the most common type of superficial (introital) dyspareunia, is a chronic pain syndrome associated with altered immune function and sensitization of nervous system.
- ⊗ **Persistent genital arousal disorder:** involves excessive genital arousal.

A disorder is diagnosed, when symptoms cause distress. Some women may not be distressed or bothered by decreased or absent sexual desire, interest, arousal or orgasm.

- ⊗ Almost all women with sexual dysfunction have features of more than one disorder. For example, chronic dyspareunia with provoked vestibulodynia (PVD) often leads to sexual desire/ interest and arousal disorders; impaired arousal may make sexual intercourse less enjoyable or even painful, decreasing the likelihood of orgasm and subsequent sexual desire. However, dyspareunia due to impaired lubrication may occur as an isolated symptom in women with a high level of sexual desire, interest and subjective arousal.
- ⊗ Female sexual disorders may be secondarily categorized as lifelong or acquired; situation specific or generalized; mild, moderate or severe based on the degree of distress it causes to the woman.
- ⊗ Although the research is limited, these disorders probably apply equally to women in heterosexual or homosexual relationships.
- ⊗ Diagnosis is done by conducting interview with both partners, separately and together; gentle pelvic examination, primarily to identify causes of dyspareunia.

Infertility-Male

Introduction

Incidence: In hospital statistics, the incidence is 10%. In general population, the incidence of infertility is 2 – 5%.

Effect of infertility: In the male dominant society of India, for infertility, the female partner is generally blamed and she becomes the victim of social gossip for barrenness. This gradually brings mental stress to the woman, later leads to psychosomatic ill health and later which may

lead to the possibility of suicide too. In most of the occasions, infertility brings forth family unhappiness, marital disharmony and ends in divorce.

Factors necessary for successful fertility

1. Testis must produce healthy sperm.
2. Sperm has to be deposited at the external os through satisfactory sexual intercourse. This has to be transported through cervical mucus, uterine tube and tubal lumen.
3. The ovary must produce healthy ovum by ovulation and ovum is picked up by the fallopian tube.
4. Fertilisation occurs at the tubal lumen and the zygote is transported through the tube to be implanted in the healthy progesterone controlled endometrium. The embryo grows to healthy baby in uterine incubator and baby is safely delivered.

Causes of infertility

1. **Systemic factors:** Age above 45 years, obesity, fatigue, heavy smoking, alcoholism
2. **Psychological factors:** nervous tension about sexual intercourse or performance anxiety, psychiatric illness causing impotency
3. **Endocrinopathy:** Hypothalamic pituitary dysfunction, hypothyroidism, diabetes mellitus
4. **Immunological causes:** Sperm auto agglutinating antibodies develop against sperm in 3-5% of the cases in vas obstruction, orchitis or testicular biopsy
5. **Coital difficulty:** Ignorance about sex, no consummation of marriage in 5% of the cases, less frequency of sexual intercourse
6. **Genital causes:**
 - a. **Defective spermatogenesis:** Undescended testis, hypospadias, Klinefelter syndrome (small firm testis), testicular damage secondary to small pox, mumps, varicocele, huge hydrocele
 - b. **Genital duct obstruction:** Herniorrhaphy repair, hydrocele repair, vas ligation, gonococcal urethritis etc.

Secondary infertility factors

- ⊗ Ageing
- ⊗ Diabetes mellitus
- ⊗ Obesity
- ⊗ Heavy smoking
- ⊗ Heavy alcoholism
- ⊗ **Endocrinopathy:** hypothyroidism, adrenal hyperplasia
- ⊗ **Psychological factors:** mental stress
- ⊗ **Genital factors:** herniorrhaphy repair, hydrocele repair, vas ligation, male reproductive systemic

infections

Investigations for male infertility: Investigations of infertility should be started within 12 months of sexual intercourse without contraception. The investigations should be completed within two years.

Before semen analysis report, ABO Rh, PPBS, urine analysis has to be done in investigation of male infertility.

Investigation of male infertility will involve 2 steps:

Step 1: Thorough clinical evaluation of husband and wife, including careful history, physical examination, examination of private parts of couple. The clinical history should include history of sexual function, sexual frequency, smoking, addition, systemic disease. It is followed by total health checkup, genital organs examination. Testicular volume is measured by orchimeter. Normal testicular volume is 90-93 ccm.

Step 2: Semen analysis of the husband: Husband goes to the standard laboratory after 3 days of abstinence, ejaculates in a wide mouth glass jar on masturbation or brings within 2 hours to the laboratory on collecting withdrawal specimen (without using condom) in a wide mouthed jar. Semen is examined after ½ an hour of collection on liquefaction.

Normal semen report	Abnormal semen report
Volume 2 – 5 ml	Aspermia – absence of semen
	Oligospermia – low volume less than 2 ml and high volume more than 5 ml
Sperm count 20 million or above/ ml	Azoospermia – 10% all infertile male – absence of sperm
Normal (20-280 million/ ml)	Oligozoospermia – sperm count less than 10 million/ ml.
	Polyzoospermia – sperm count more than 250 million/ ml

Sperm morphology

Normal sperms 60% or above	
Sperm motility – 60% one hour; 50% 2 hours	
	Grade 0 – no motion (necrozoospermia)
Grade 3 – Good; Forward progression with good speed	Grade 1 – Poor; weak or sluggish forward progression of sperm
Grade 4 – Excellent – vigorous rapid forward movement	Grade 2 – Moderate; Definite forward progression movement
Pus cells – 1-5 / high power field	Pus cells > 5/ high power field indicates infection of genital tract

Normal semen report	Abnormal semen report
Fructose content – derived from seminal vesicle; normal 120-180 mg/ dl; this is not routinely tested	Fructose content is absent in azoospermia which helps to identify seminal duct obstruction where fructose content becomes absent.

On normal semen report, 3 report, no repeat semen test becomes necessary

On abnormal semen report, 3 semen testing at monthly interval is needed from standard laboratory.

On report of azoospermia, investigations on wife is suspended.

Female Infertility

Introduction:

There are two terms used in the field of infertility, infertility and sterility. Both literally mean inability to conceive after one year of sexual life without contraception when the couple are anxious to have progeny. Some distinguish the two terms with slight difference. Sterility refers to the absolute state of inability to conceive while infertility refers to the relative state of failure to conceive. Infertility can be either primary, where the conception has never occurred or secondary, where conception has failed to occur after a period of fertility.

Incidence: In hospital statistics, the incidence is 10%. In general population, the incidence of infertility is 2 – 5%.

Effect of infertility: In the male dominant society of India, for infertility, the female partner is generally blamed and she becomes the victim of social gossip for barrenness. This gradually brings mental stress to the woman, later leads to psychosomatic illhealth and later which may lead to the possibility of suicide too. In most of the occasions, infertility brings forth family unhappiness, marital disharmony and ends in divorce.

Factors necessary for successful fertility

1. Testis must produce healthy sperm.
2. Sperm has to be deposited at the external os through satisfactory sexual intercourse. This has to be transported through cervical mucus, uterine tube and tubal lumen.
3. The ovary must produce healthy ovum by ovulation and ovum is picked up by the fallopian tube.
4. Fertilisation occurs at the tubal lumen and the zygote is transported through the tube to be implanted in the healthy progesterone controlled endometrium. The embryo grows to healthy baby in uterine incubator and baby is safely delivered.

One or multiple factors of the above mechanisms are responsible for the complaint of infertility. It could be either physiological or pathological.

- ⊗ **Physiological:** Infertility is present before puberty, after menopause, during pregnancy and sometimes during lactation. During these periods, ovulation does not occur. Physiological sterility exists in a couple who have contracted marriage before or after the reproductive age.
- ⊗ **Pathological:**
 - Faulty male factors responsible for 25% of the cases.
 - Faulty female factors responsible for 50% of the cases.
 - Faulty combined factors responsible for another 25% of the cases.
- ⊗ In about 10% of the cases, no obvious abnormality can be found to explain the exact cause of infertility even when current diagnostic facilities such as endoscopy, endocrine assays are available.

Female factors of infertility:

1. **Systemic factors:** Age over 35 years, obesity. May women conceive easily with pulmonary tuberculosis, severe anemia in India.
2. **Psychological factors:** Fear of performing sexual intercourse, marital disharmony, work through tubal spasm and infrequent sexual intercourse
3. **Endocrinopathy:** Thyroid disorders, hypogonadotrophic amenorrhoea, metropathic bleeding, PCOD
4. **Immunological incompatibility:** Agglutinating antibodies against sperm present in cervical mucus or in woman's serum. ABO antigen may agglutinate in cervical mucus.
5. **Genital factors:**
 - a. **Vaginal factors** – 2% of the cases, e.g. narrow vaginal orifice, vaginismus, vaginitis, vaginal atresia or stenosis, escape of semen after coitus (fluor seminis)
 - b. **Cervical factors** – 20% of the cases
 - i. **Hormonal** – good oestrogen effect causes positive fern test and formation of spinnbarkeit.
 - ii. **Anatomical defect** – elongation, intracervical obstruction, cervical scarring
 - iii. **Cervicitis** – with poor mucus and pus cells due to pyogenic or tubercular infection
 - iv. **Neoplastic** – cervical polyp
 - v. **Immunological** – sperm agglutinating antibodies in cervical mucus or woman's serum may hamper sperm motility
 - c. **Uterine factors** – 10% of the cases – uterine fibromyoma, adenomyosis, uterine hyperplasia,

infantile uterus, tubercular endometritis

- d. **Tubal factors** – 10-40% of the cases, tubal occlusions or peritubal adhesions caused by gonococcal, pyogenic, tubercular or chlamydial infections
- e. **Ovarian factors** – 5% of the cases caused by anovulation. Hyperprolactinemia and hypothyroidism are the probable causes of female infertility.
- f. **Peritoneal factors** – peritubal adhesion, pelvic endometriosis

Combined male and female factors: Most infertility problems will arise due to

1. Inadequate sperm production
2. Failure of ovulation
3. Abnormalities in female reproductive tract

Secondary infertility factors:

- ⊗ Ageing
- ⊗ Obesity
- ⊗ **Endocrinopathy:** Thyroid disturbances, metropathia, Sheehan's syndrome
- ⊗ Vaginal stenosis, cervical stenosis, prolapse, endometrial tuberculosis
- ⊗ Uterine fibroid
- ⊗ Ovarian tumours
- ⊗ Pelvic endometriosis
- ⊗ Posttubal sterilisation

Investigation of infertility:

When to investigate? 90% of the women conceive by 18 months of sexual exposure without contraception. Therefore, investigation is performed when

- ⊗ Woman below 25 years remains infertile after 2 years of normal sexual life
- ⊗ Woman of 25 to 35 years remains infertile after 1 year of normal sexual life
- ⊗ Woman of above 35 years remains infertile after 6 months of normal sexual life

For those who are unnecessarily worried before adequate period of sexual exposure, clinical examination of the couple and assurance with explanation of physiology of conception are all that are necessary. Women with severe heart disease, chronic kidney disease, severe hypertension and diabetes mellitus should be examined accurately and they should be discouraged to conceive unless opinion of physician is taken.

Duration of investigations:

A thorough diagnostic workup should identify one or more causes of infertility in about 90% of the couples.

The investigations should be completed within 2 years through several consultations.

Investigation of infertility includes 7 steps

1. **Step 1:** Clinical history of both husband and wife, proper evaluation including physical examination, examination of private parts of the couple
2. **Step 2:** Semen analysis of the husband: The husband goes to a standard laboratory after 3 days of abstinence, ejaculates in wide mouth jar on masturbation or brings within 2 hours to laboratory on collecting withdrawal specimen of semen without using the condom in a wide mouthed jar. Semen is examined after ½ an hour of collection on liquefaction.

Routine testing of wife: Blood Hb, ABO Rh, VDRL, PPBS and urine analysis

Routine testing of husband: ABO Rh, VDRL, PPBS, urine analysis

3. **Step 3:** Cervical dilatation and curettage (D and C) under local or general anesthesia. This is performed during 18-21st day of menstrual cycle to remove intracervical obstruction; to identify secretory endometrium and to exclude endometrial tuberculosis.
4. **Step 4: Tubal patency tests:** Hysterosalpingography (HSG): This is done after 6 months of D and C, if no pregnancy occurs. This is done on radiographic table within a week following end of menses by introducing water soluble radio opaque dye into the uterine cavity through a canula. This helps in identifying tubal patency, tubal luminal block and hydrosalpinx.
5. **Step 5: Ovulation tests:** These are done after 6 months of HSG. No direct procedure is available. Urine pregnancy test is the surest proof.

Basal Body Temperature (BBT) test: Oral temperature is recorded for one minute continuously in a graph or table for 3-4 cycles starting from the first day of menstruation. She takes oral temperature by a clinical thermometer on waking before micturition.

• **Ovulatory BBT** is biphasic sharp drop of 0.5°F temperature followed by sustained rise by 1°F during luteal phase.

• **Anovulatory BBT** – monophasic BBT with no rise of temperature during the luteal phase of menstrual cycle.

6. Step 6: Tests for cervical factors:

• **Fern test:** of cervical mucus is an indirect evidence of ovulation showing palm leaf or fern like pattern of cervical mucus under microscope when mucus is dried on a slide. This results from crystallisation of sodium chloride with mucus due to oestrogen effect. Cervical mucus is collected from cervical canal by uterine dressing forceps. Mucus is smeared on a glass slide (washed with distilled water) and dried for 15 minutes and examined under high or low power microscope.

➤ Ferning of cervical mucus starts at 5th or 6th day of menstrual cycle, becomes maximum 24-48 hours before ovulation and is replaced by beaded cellular pattern under progesterone effect 24-72 hours after ovulation and is absent completely after 22nd day of cycle. Therefore, presence of ferning by 22nd day indicates anovulation.

• **Spinnbarkeit (Thread test, Rheology test):** Ovulation mucus has the property of elasticity and withstands stretching into a thread at least 6 cms. This is tested by pulling apart the cover slip placed on drop of cervical mucus (oestrogenic effect). Mucus becomes viscous, the thread fractures when put under tension during luteal phase.

7. **Step 7: Laparoscopic dye test:** is not routinely done. It is indicated in abnormal HSG findings and in cases where all above steps show normalcy.

Double puncture laparoscopy: This is performed within a week after menstruation. Methylene blue dye is pushed through HSG canula into uterine cavity. Laparoscopy shows spillage of dye through fimbrial end or no spillage secondary to cornual block or spasm.

Key summary of the chapter

- Sexual dysfunction and infertility are two different symptoms that can disturb the emotional or mental peace of the couple; each of which are having distinctive implications. Sexual dysfunction may disturb the couple only; whereas couple; each of which are having distinctive implications. Sexual dysfunction may disturb the couple only; whereas presenting complaint of infertility may disturb the couple in association with people who are associated with these couple and thus leading to the social disturbances too. Earlier, discussion in related to sexual dysfunction and infertility were considered to be a taboo, but now because of excess information available in the social media, internet and peer influences.
- **Male sexual function involves** intact libido (sexual desire), ability to achieve and maintain erection, ejaculation and orgasm and Detumescence.
- **Female sexual function involves** intact libido (sexual desire), sexual arousal due to environment and partner, vaso congestion and lubrication of vagina and external genitalia and orgasm.
- In case of male and female sexual dysfunction, any one or two or more of the categories of symptoms will be observed in the male or female which in turn disturb the couple. **Erectile dysfunction (ED)** is the most common male sexual

dysfunction noticed in males and in females, decreased sexual responsiveness (sexual desire/interest disorder, sexual arousal disorder, orgasmic disorder, vaginismus and dyspareunia) and increased sexual responsiveness (persistent genital arousal disorder) are observed.

- ⊗ Infertility is defined as inability of the couple to have conception even after one year of unprotected sexual intercourse. There are two terms used in the field of infertility, infertility and sterility. Both literally mean inability to conceive after one year of sexual life without contraception when the couple are anxious to have progeny. Some distinguish the two terms with slight difference. Sterility refers to the absolute state of inability to conceive while infertility refers to the relative state of failure to conceive.
- ⊗ Infertility can be physiological or pathological. Physiological infertility is seen in females before puberty, after menopause, during pregnancy and sometimes during lactation. This can be observed in a couple who have contracted marriage before or after the reproductive age. Whereas, pathological infertility is seen in males in 25% of the cases; in females in about 50% of the cases and in both genders in another 25% of the cases. Among 100% of the cases of infertility, 10% of the cases will have no obvious abnormality can be found to explain the exact cause of infertility even when current diagnostic facilities such as endoscopy, endocrine assays are available.
- ⊗ While doing the investigations of infertility in a couple, first of all, investigation of male is being carried out. If semen analysis is showing significant abnormal changes, then investigation of female partner is suspended. Investigation of male partner involves 2 steps process, whereas investigation of the female partner involves 7 steps process in the evaluation of the infertility.

MCQs

1. Normal volume of semen is

- a. 0.5 to 3 ml; b. 2 to 5 ml;
- c. 5 to 10 ml; d. 10 to 15 ml

Answer: (b)

2. The term dyspareunia refers to pain observed during

- a. Menstruation;
- b. Sexual intercourse;
- c. Unrelated to menstruation;
- d. Menopause

Answer: (b)

3. Prostatic fluid contains which of the following?

- a. Glucose; b. Lactose;
- c. Fructose; d. Galactose

Answer: (c)

4. Fertilisation of zygote occurs in which of the following part of the female reproductive system?

- a. Vulva; b. Vagina;
- c. Uterus; d. Uterine tube

Answer: (d)

5. D and C (dilatation and curettage) is performed duringdays of menstrual cycle.

- a. 10 to 12th day; b. 22 to 25th day;
- c. During menstruation; d. 18 to 21st day

Answer: (d)

6. To perform Fern test, mucus of which of the following part is used?

- a. Vagina; b. Uterine cavity;
- c. Fallopian tube; d. Cervix

Answer: (d)

7. In basal body temperature test, temperature of the patient is taken

- a. Before micturition;
- b. After micturition;

- c. Before defaecation;
- d. No such time period is specified

Answer: (a)

8. Laparoscopic dye test is done in which step of evaluation of female infertility?

- a. Step 2; b. Step 6;
- c. Step 7; Step 1

Answer: (c)

9. The part involved in hypogonadotropic amenorrhea is

- a. Thyroid gland; b. Ovary;
- c. Adrenal gland; d. Pituitary gland

Answer: (d)

10. The term oligozoospermia refers to the sperm count less than

- a. 200 millions/ ml; b. 50 millions/ ml;
- c. 5 millions/ ml; d. 10 millions/ ml

Answer: (d)

Chapter resources

- ⊗ <https://psychicare.com/sexual-dysfunction/>
- ⊗ The Merck Manual of Diagnosis and Therapy by Robert S Porter and Justin L Kaplan, 19th edition, Merck Sharp & Dohme Corp, New Jersey, 2011
- ⊗ Golwalla's medicine for students by Aspi F Golwalla and Sharukh A Golwalla, edited by Milind Y Nadkar, 25th edition, Jaypee Brothers Medical Publishers (P) Ltd, New Delhi, 2017
- ⊗ Chamberlain's symptoms and signs in clinical medicine by Colin Ogilvie and Christopher C Evans, 12th edition, Butterworth-Heinemann Publishers, 1997
- ⊗ Best & Taylor's physiological basis of medical practice by O P Tandon and Y Tripathi, 13th edition, Wolters Kluwer Health (India) Pvt. Ltd, Gurgaon, 2012
- ⊗ Textbook of Medical physiology by Arthur C Guyton and John E Hall, 11th edition, Saunders Elsevier publications, Philadelphia, 2006
- ⊗ C S Dawn. Undergraduate and postgraduate textbook of gynaecology and contraception, 13th edition, Dawn Books, Calcutta, 1999.